GCSE Biology
Exam Walkthrough Pack

Step-by-step worked exam-style questions with full explanations

Original JDScience exam-style questions — not copied from past papers

Introduction

This Biology Exam Walkthrough Pack is designed to help you understand how to approach GCSE and IGCSE exam-style questions. Every question in this booklet is an original JDScience item written to match common specification skills — it is not copied from AQA, Edexcel, OCR or other past papers.

How to use this pack: read each question, cover the walkthrough, and try answering in exam conditions first. Then read what the question is asking, follow the step-by-step thinking, compare your answer with the model answer, and study the mark breakdown to see exactly how marks are awarded.

Command words tell you what type of answer is required. Section 1 explains the most common GCSE science command words. Always underline the command word before you start writing.

Showing working matters: in calculations, examiners award method marks even when the final answer is wrong. In longer answers, a clear logical sequence helps you secure every available mark.

Examiners award marks for correct science linked to the question. Keywords from the specification matter, but a correct idea in your own words can still score. Extended questions often use level-based marking: plan before you write, use paragraphs, and link ideas.

To move from Grade 4/5 to Grade 7/8/9: use precise terminology, explain mechanisms (not just names), include data or examples where asked, and check units and significant figures in calculations. For six-mark questions, a short plan and linked paragraphs beat a long unordered list.

Section 1: Command word guide

State

What you must do: Give a short, factual answer — often one word, number or phrase. No explanation is required unless the question also asks you to explain.

Common mistake: Writing a long paragraph when a single fact is enough, or giving an explanation when only a fact was asked for.

Model sentence starter: The [quantity/name] is …

Give

What you must do: Provide the information requested. Similar to state, but may need a short phrase or named example.

Common mistake: Listing unrelated facts that do not answer the question.

Model sentence starter: One example is … / The value is …

Describe

What you must do: Say what happens or what something is like. Focus on observations, patterns or features. You do not need to say why.

Common mistake: Explaining causes when the command word is only describe.

Model sentence starter: First … then … / The pattern shows …

Explain

What you must do: Give reasons why something happens. Link cause and effect using science ideas and key terms.

Common mistake: Describing what happens without saying why, or using everyday language instead of science.

Model sentence starter: This happens because … which means …

Calculate

What you must do: Work out a numerical answer. Show your working, include units, and give the final value to the required precision.

Common mistake: Missing units, no working, or using the wrong equation.

Model sentence starter: Use … = … / Substitute: …

Compare

What you must do: Identify similarities and/or differences between two or more things. Refer to both sides.

Common mistake: Only describing one item, or listing features without comparing.

Model sentence starter: Both … however … / X has … whereas Y …

Evaluate

What you must do: Weigh up evidence or options, consider strengths and weaknesses, and reach a supported judgement.

Common mistake: Only listing advantages with no conclusion or balanced comment.

Model sentence starter: An advantage is … A limitation is … Overall …

Suggest

What you must do: Apply your knowledge to a new situation. Your answer should be sensible and scientifically plausible.

Common mistake: Repeating textbook facts that do not fit the context given.

Model sentence starter: A possible reason is … / This could be because …

Justify

What you must do: Give evidence or reasons that support a choice, conclusion or statement already made.

Common mistake: Restating the claim without giving supporting science.

Model sentence starter: This is supported by … because …

Determine

What you must do: Find out a value or conclusion using data, a graph or a calculation. Show how you reached your answer.

Common mistake: Reading a graph incorrectly or not showing how the value was obtained.

Model sentence starter: From the graph … / Using the data …

Analyse

What you must do: Break information into parts, interpret patterns or trends, and explain what the data shows.

Common mistake: Describing data without interpreting it, or ignoring anomalies.

Model sentence starter: The data shows that … / As X increases, Y …

Section 2: Cell biology

QUESTION 1

Name the organelle that controls the activities of the cell.

QUESTION TYPE

Topic: Cell biology | **Command word:** state | **Marks:** 1 | **Skill tested:** Cell structure

WHAT THE QUESTION IS ASKING

Identify the control centre of the cell.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The nucleus.

MARK BREAKDOWN

Mark 1: The nucleus.

COMMON MISTAKES

- Writing mitochondrion.
- Giving a description instead of a name.

EXAMINER TIP

For one-mark recall, give a single precise term.

QUESTION 2

State the function of ribosomes.

QUESTION TYPE

Topic: Cell biology | **Command word:** state | **Marks:** 1 | **Skill tested:** Cell structure

WHAT THE QUESTION IS ASKING

State what ribosomes do.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Protein synthesis.

MARK BREAKDOWN

Mark 1: Protein synthesis.

COMMON MISTAKES

- Saying respiration.
- Saying they make DNA.

EXAMINER TIP

Link ribosomes to proteins, not energy.

QUESTION 3

Give the name of the structure that makes a plant cell rigid.

QUESTION TYPE

Topic: Cell biology | **Command word:** give | **Marks:** 1 | **Skill tested:** Plant cells

WHAT THE QUESTION IS ASKING

Name the plant cell wall component.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Cell wall (made of cellulose).

MARK BREAKDOWN

Mark 1: Cell wall (made of cellulose).

COMMON MISTAKES

- Saying membrane.
- Saying vacuole.

EXAMINER TIP

Cell wall is outside the membrane in plant cells.

QUESTION 4

Describe two differences between a prokaryotic cell and a eukaryotic cell.

QUESTION TYPE

Topic: Cell biology | **Command word:** describe | **Marks:** 2 | **Skill tested:** Cell types

WHAT THE QUESTION IS ASKING

Compare prokaryotes and eukaryotes structurally.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Prokaryotic cells have no nucleus — DNA is free in the cytoplasm — and are smaller. Eukaryotic cells have a membrane-bound nucleus and membrane-bound organelles.

MARK BREAKDOWN

Mark 1: One valid structural difference.

Mark 2: Second valid structural difference.

COMMON MISTAKES

- Only describing one type.
- Stating functions without structural differences.

EXAMINER TIP

Nucleus presence and size are reliable two-mark pairs.

QUESTION 5

Explain why root hair cells have many mitochondria.

QUESTION TYPE

Topic: Cell biology | **Command word:** explain | **Marks:** 2 | **Skill tested:** Specialised cells

WHAT THE QUESTION IS ASKING

Link structure to function for active transport.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Root hair cells absorb mineral ions by active transport, which requires ATP from aerobic respiration in mitochondria.

MARK BREAKDOWN

Mark 1: Active transport or ion uptake mentioned.

Mark 2: Link to ATP or respiration in mitochondria.

COMMON MISTAKES

- Saying photosynthesis.
- Not linking mitochondria to energy need.

EXAMINER TIP

Always connect organelle to the process that needs energy.

QUESTION 6

Compare diffusion and osmosis.

QUESTION TYPE

Topic: Cell biology | **Command word:** compare | **Marks:** 2 | **Skill tested:** Transport

WHAT THE QUESTION IS ASKING

State a similarity and a difference.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Both involve passive movement down a concentration gradient without energy. Diffusion can involve any particle; osmosis is the movement of water only across a partially permeable membrane.

MARK BREAKDOWN

Mark 1: Valid similarity.

Mark 2: Valid difference about water or membrane.

COMMON MISTAKES

- Saying osmosis needs energy.
- Not mentioning water specifically.

EXAMINER TIP

Use 'both … however …' structure.

QUESTION 7

Explain how the structure of the cell-surface membrane helps it control what enters and leaves the cell.

QUESTION TYPE

Topic: Cell biology | **Command word:** explain | **Marks:** 3 | **Skill tested:** Membranes

WHAT THE QUESTION IS ASKING

Link phospholipid bilayer and proteins to selective permeability.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The membrane is a phospholipid bilayer forming a barrier to many polar molecules. Embedded proteins act as channels or carriers for specific substances, so only selected ions and molecules can cross.

MARK BREAKDOWN

Mark 1: Phospholipid bilayer or barrier idea.
Mark 2: Proteins as channels or carriers.
Mark 3: Selective control of substances.

COMMON MISTAKES

- Calling the membrane fully permeable.
- Ignoring protein role.

EXAMINER TIP

Mention both the lipid barrier and protein channels.

QUESTION 8

Describe the stages of mitosis in order.

QUESTION TYPE

Topic: Cell biology | **Command word:** describe | **Marks:** 3 | **Skill tested:** Cell division

WHAT THE QUESTION IS ASKING

Outline the sequence of mitosis.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Chromosomes condense and become visible. They line up at the equator of the cell. Sister chromatids are pulled to opposite poles. Two genetically identical nuclei form before cytokinesis.

MARK BREAKDOWN

Mark 1: Condensation or visibility.

Mark 2: Alignment at equator.

Mark 3: Separation to poles and two nuclei.

COMMON MISTAKES

- Mixing mitosis with meiosis.
- Missing the order.

EXAMINER TIP

Learn PMAT: prophase, metaphase, anaphase, telophase.

QUESTION 9

A student claims plant cells cannot respire because they contain chloroplasts. Evaluate this statement.

QUESTION TYPE

Topic: Cell biology | **Command word:** evaluate | **Marks:** 4 | **Skill tested:** Metabolism

WHAT THE QUESTION IS ASKING

Judge the claim using plant cell metabolism.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The statement is incorrect. Chloroplasts carry out photosynthesis, but mitochondria in plant cells carry out aerobic respiration at all times, releasing ATP for growth and active transport. At night, when photosynthesis stops, respiration continues. Both processes occur in plant cells.

MARK BREAKDOWN

Mark 1: Statement judged incorrect.

Mark 2: Photosynthesis in chloroplasts stated.

Mark 3: Respiration in mitochondria stated.

Mark 4: Balanced conclusion with day/night or both organelles.

COMMON MISTAKES

- Agreeing with the student.
- Saying plants only photosynthesise.

EXAMINER TIP

Plants respire continuously; they photosynthesise only in light.

QUESTION 10

Explain why multicellular organisms need specialised cells and differentiation.

QUESTION TYPE

Topic: Cell biology | **Command word:** explain | **Marks:** 4 | **Skill tested:** Specialisation

WHAT THE QUESTION IS ASKING

Link specialisation to efficiency in large organisms.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Differentiation produces cells with specific structures for particular jobs, such as muscle cells for contraction or red blood cells for oxygen transport. This division of labour is more efficient than every cell performing every function, allowing complex tissues and organs to form.

MARK BREAKDOWN

Mark 1: Differentiation defined or implied.

Mark 2: Example of specialised cell.

Mark 3: Division of labour idea.

Mark 4: Efficiency or complexity benefit.

COMMON MISTAKES

- Confusing with evolution.
- No example given.

EXAMINER TIP

Give one named specialised cell as evidence.

QUESTION 11

Describe how you would prepare a microscope slide of onion epidermis and use it to estimate cell size. Include calibration, repeat measurements and how you would improve reliability.

QUESTION TYPE

Topic: Cell biology | **Command word:** describe | **Marks:** 6 | **Skill tested:** Microscopy

WHAT THE QUESTION IS ASKING

Full microscopy method with size calculation and evaluation.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Peel a thin layer of epidermis from an onion, place on a slide with a drop of iodine stain and a coverslip, avoiding air bubbles. View under low then high power, focusing with fine adjustment. Calibrate an eyepiece graticule against a stage micrometer to measure cell length. Repeat measurements on ten cells and calculate a mean. To improve reliability, repeat with a second onion bulb and use the same magnification each time.

MARK BREAKDOWN

Mark 1: Valid slide preparation.
Mark 2: Microscope use including focusing.
Mark 3: Graticule or micrometer calibration.
Mark 4: Repeat measurements and mean.
Mark 5: Reliability improvement.
Mark 6: Clear logical sequence.

COMMON MISTAKES

- No stain or coverslip.
- Skipping calibration.
- No repeat or mean.

EXAMINER TIP

Six-mark practical answers need method, measurement and improvement.

GRADE 7–9 EXTENSION

Reference eyepiece × objective = total magnification.

QUESTION 12

An electron micrograph shows a mitochondrion 8 mm long. The actual length is 4 µm. Calculate the magnification. Show your working.

QUESTION TYPE

Topic: Cell biology | **Command word:** calculate | **Marks:** 3 | **Skill tested:** Calculation

WHAT THE QUESTION IS ASKING

Use magnification = image size ÷ actual size with consistent units.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Magnification = image size ÷ actual size = 8 mm ÷ 0.004 mm = ×2000.

MARK BREAKDOWN

Mark 1: Correct formula stated.
Mark 2: Unit conversion (4 µm = 0.004 mm).
Mark 3: Correct final magnification.

COMMON MISTAKES

- Not converting µm to mm.
- Dividing actual by image.

EXAMINER TIP

Convert both measurements to the same unit before dividing.

QUESTION 13

During a required practical on osmosis in potato cylinders, describe how you would identify the independent, dependent and control variables, and one source of random error.

QUESTION TYPE

Topic: Cell biology | **Command word:** describe | **Marks:** 4 | **Skill tested:** Required practical — osmosis

WHAT THE QUESTION IS ASKING

Apply variables framework to osmosis investigation.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Independent variable: concentration of sugar solution. Dependent variable: change in mass or length of potato cylinders. Control variables: same potato, same cylinder dimensions, same immersion time and temperature. A random error is slight variation in initial mass due to different water content in different parts of the potato.

MARK BREAKDOWN

Mark 1: Correct independent variable.

Mark 2: Correct dependent variable.

Mark 3: One control variable.

Mark 4: Valid random error.

COMMON MISTAKES

- Swapping independent and dependent.
- Calling temperature independent when it should be controlled.

EXAMINER TIP

What you change → independent; what you measure → dependent.

Section 2: Organisation

QUESTION 14

Name the organ that produces bile.

QUESTION TYPE

Topic: Organisation | **Command word:** state | **Marks:** 1 | **Skill tested:** Digestive system

WHAT THE QUESTION IS ASKING

Recall the source of bile.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The liver.

MARK BREAKDOWN

Mark 1: The liver.

COMMON MISTAKES

- Pancreas.
- Stomach.

EXAMINER TIP

Bile is made in the liver and stored in the gall bladder.

QUESTION 15

State the function of the alveoli in the lungs.

QUESTION TYPE

Topic: Organisation | **Command word:** state | **Marks:** 1 | **Skill tested:** Gas exchange

WHAT THE QUESTION IS ASKING

State what alveoli do.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Gas exchange (oxygen into blood and carbon dioxide out).

MARK BREAKDOWN

Mark 1: Gas exchange (oxygen into blood and carbon dioxide out).

COMMON MISTAKES

- Breathing only.
- Making mucus.

EXAMINER TIP

Link alveoli to diffusion of gases.

QUESTION 16

Give the name of the blood vessel that carries blood away from the heart.

QUESTION TYPE

Topic: Organisation | **Command word:** give | **Marks:** 1 | **Skill tested:** Circulatory system

WHAT THE QUESTION IS ASKING

Name the vessel type leaving the heart.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Artery.

MARK BREAKDOWN

Mark 1: Artery.

COMMON MISTAKES

- Vein.
- Capillary.

EXAMINER TIP

Arteries carry blood away; veins carry blood towards the heart.

QUESTION 17

Describe the function of xylem in a plant.

QUESTION TYPE

Topic: Organisation | **Command word:** describe | **Marks:** 2 | **Skill tested:** Plant transport

WHAT THE QUESTION IS ASKING

Explain water and mineral transport role.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Xylem transports water and mineral ions from roots to leaves. It is made of dead hollow cells strengthened with lignin.

MARK BREAKDOWN

Mark 1: Transports water and minerals upward.

Mark 2: Dead hollow cells or lignin mentioned.

COMMON MISTAKES

- Saying xylem carries sugars.
- Confusing with phloem.

EXAMINER TIP

Xylem = water up; phloem = sugars both ways.

QUESTION 18

Explain why the small intestine has a large surface area.

QUESTION TYPE

Topic: Organisation | **Command word:** explain | **Marks:** 2 | **Skill tested:** Digestion

WHAT THE QUESTION IS ASKING

Link villi and microvilli to absorption rate.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

A large surface area increases the rate of absorption of digested food molecules into the blood by diffusion and active transport.

MARK BREAKDOWN

Mark 1: Large surface area stated.

Mark 2: Link to faster absorption or more diffusion.

COMMON MISTAKES

- Saying digestion happens faster only.
- No link to absorption.

EXAMINER TIP

Surface area always links to rate of exchange.

QUESTION 19

Compare arteries and veins.

QUESTION TYPE

Topic: Organisation | **Command word:** compare | **Marks:** 2 | **Skill tested:** Circulatory system

WHAT THE QUESTION IS ASKING

State one similarity and one difference.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Both carry blood and have a lumen. Arteries have thick muscular walls and carry blood at high pressure away from the heart; veins have thinner walls, valves, and carry blood at lower pressure towards the heart.

MARK BREAKDOWN

Mark 1: Valid similarity.

Mark 2: Valid difference in wall, pressure or direction.

COMMON MISTAKES

- Only describing arteries.
- Saying veins carry oxygenated blood only.

EXAMINER TIP

Mention wall thickness or valves for a strong difference mark.

QUESTION 20

Explain how the structure of a red blood cell is adapted to its function.

QUESTION TYPE

Topic: Organisation | **Command word:** explain | **Marks:** 3 | **Skill tested:** Blood

WHAT THE QUESTION IS ASKING

Link biconcave shape, haemoglobin and no nucleus to oxygen transport.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Red blood cells have no nucleus to leave more room for haemoglobin, which binds oxygen. The biconcave shape increases surface area for faster diffusion of oxygen. They are small and flexible to pass through capillaries.

MARK BREAKDOWN

Mark 1: No nucleus or more haemoglobin.
Mark 2: Haemoglobin binds oxygen.
Mark 3: Biconcave shape increases surface area.

COMMON MISTAKES

- Saying they have a nucleus.
- Missing haemoglobin.

EXAMINER TIP

Structure–function answers need named feature plus reason.

QUESTION 21

Describe how enzymes work in digestion.

QUESTION TYPE

Topic: Organisation | **Command word:** describe | **Marks:** 3 | **Skill tested:** Enzymes

WHAT THE QUESTION IS ASKING

Outline lock-and-key and where enzymes act.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Enzymes are biological catalysts with active sites shaped for specific substrates. Amylase in the mouth and small intestine breaks starch to sugars; proteases break proteins to amino acids; lipase breaks lipids to fatty acids and glycerol.

MARK BREAKDOWN

Mark 1: Enzyme–substrase specificity or active site.

Mark 2: Named digestive enzyme and substrate.

Mark 3: Products of digestion stated.

COMMON MISTAKES

- Saying enzymes are used up.
- No named enzyme.

EXAMINER TIP

Name at least one enzyme, substrate and product.

QUESTION 22

A patient has narrowed coronary arteries. Explain how this could affect the body.

QUESTION TYPE

Topic: Organisation | **Command word:** explain | **Marks:** 4 | **Skill tested:** Disease

WHAT THE QUESTION IS ASKING

Link reduced blood flow to heart muscle oxygen and respiration.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Narrowed coronary arteries reduce blood flow to heart muscle. Less oxygen and glucose reach cardiac muscle cells, so aerobic respiration decreases and less ATP is available for contraction. This can cause chest pain or heart attack.

MARK BREAKDOWN

Mark 1: Reduced blood flow to heart muscle.

Mark 2: Less oxygen or glucose.

Mark 3: Less aerobic respiration or ATP.

Mark 4: Effect on contraction or health.

COMMON MISTAKES

- Confusing with lung disease.
- No link to respiration.

EXAMINER TIP

Follow the chain: artery → oxygen → respiration → muscle function.

QUESTION 23

Describe how translocation occurs in plants.

QUESTION TYPE

Topic: Organisation | **Command word:** describe | **Marks:** 4 | **Skill tested:** Plant transport

WHAT THE QUESTION IS ASKING

Explain movement of sugars in phloem.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Dissolved sugars are loaded into phloem sieve tubes in sources such as leaves. Water follows by osmosis, creating pressure that moves sap to sinks such as roots or fruits, where sugars are unloaded for storage or use.

MARK BREAKDOWN

Mark 1: Sugars in phloem.
Mark 2: Source to sink direction.
Mark 3: Osmosis or pressure flow idea.
Mark 4: Unload at sink.

COMMON MISTAKES

- Saying xylem transports sugar.
- No source/sink idea.

EXAMINER TIP

Phloem moves sugars; xylem moves water.

QUESTION 24

Describe how you would investigate the effect of light intensity on the rate of transpiration using a potometer. Include variables, measurements and how to improve accuracy.

QUESTION TYPE

Topic: Organisation | **Command word:** describe | **Marks:** 6 | **Skill tested:** Transpiration practical

WHAT THE QUESTION IS ASKING

Full method for potometer investigation with evaluation.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Cut a leafy shoot under water and attach to a filled potometer with no air bubbles. Measure distance the water meniscus moves in a set time at low light. Repeat at higher light intensity, keeping temperature and humidity constant. Calculate rate from distance ÷ time. Repeat entire setup and calculate mean. Improve accuracy by using a longer capillary tube for larger readings or a data logger.

MARK BREAKDOWN

Mark 1: Potometer set up without air bubbles.

Mark 2: Light intensity changed.

Mark 3: Rate calculated from distance and time.

Mark 4: Control variables named.

Mark 5: Repeat and mean.

Mark 6: Accuracy improvement.

COMMON MISTAKES

- No time measurement.
- Changing temperature and light together.

EXAMINER TIP

State what you measure and how you calculate rate.

GRADE 7–9 EXTENSION

Link increased transpiration to faster water uptake at higher light.

QUESTION 25

A student calculates the mean length of five potato cylinders as 4.2 cm. The lengths are 4.0, 4.1, 4.3, 4.5 and 4.1 cm. Calculate the range.

QUESTION TYPE

Topic: Organisation | **Command word:** calculate | **Marks:** 3 | **Skill tested:** Calculation

WHAT THE QUESTION IS ASKING

Range = highest value – lowest value.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Range = 4.5 – 4.0 = 0.5 cm.

MARK BREAKDOWN

Mark 1: Correct formula or method stated.

Mark 2: Identifies highest and lowest.

Mark 3: Range = 0.5 cm.

COMMON MISTAKES

- Using mean in the calculation.
- Wrong min or max.

EXAMINER TIP

Range shows spread; it is not the mean.

QUESTION 26

Describe how you would test a food sample for starch and for protein using standard reagents.

QUESTION TYPE

Topic: Organisation | **Command word:** describe | **Marks:** 4 | **Skill tested:** Required practical — food tests

WHAT THE QUESTION IS ASKING

Outline iodine and Biuret tests with positive results.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Add a few drops of iodine solution to the sample; a blue-black colour indicates starch. For protein, add Biuret reagent and mix; a purple/lilac colour indicates protein. Use fresh samples and record colours against controls with no starch or protein.

MARK BREAKDOWN

Mark 1: Iodine test with blue-black for starch.

Mark 2: Biuret test with purple for protein.

Mark 3: Control or comparison mentioned.

Mark 4: Clear method order.

COMMON MISTAKES

- Wrong colours.
- Using Benedict's for protein.

EXAMINER TIP

Learn colour changes: iodine blue-black, Biuret purple.

Section 2: Infection and response

QUESTION 27

State what is meant by a pathogen.

QUESTION TYPE

Topic: Infection and response | **Command word:** state | **Marks:** 1 | **Skill tested:** Pathogens

WHAT THE QUESTION IS ASKING

Define disease-causing organism.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

A microorganism that causes disease.

MARK BREAKDOWN

Mark 1: A microorganism that causes disease.

COMMON MISTAKES

- Any microbe regardless of harm.
- A toxin only.

EXAMINER TIP

Pathogens cause infectious disease.

QUESTION 28

Give one example of a bacterial disease.

QUESTION TYPE

Topic: Infection and response | **Command word:** give | **Marks:** 1 | **Skill tested:** Disease

WHAT THE QUESTION IS ASKING

Name a disease caused by bacteria.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Salmonella food poisoning (or tuberculosis, or gonorrhoea).

MARK BREAKDOWN

Mark 1: Salmonella food poisoning (or tuberculosis, or gonorrhoea).

COMMON MISTAKES

- Naming a viral disease.
- Vague 'infection' without disease name.

EXAMINER TIP

Have one bacterial and one viral example ready.

QUESTION 29

Name the type of cell that engulfs pathogens as part of the immune response.

QUESTION TYPE

Topic: Infection and response | **Command word:** state | **Marks:** 1 | **Skill tested:** Immune system

WHAT THE QUESTION IS ASKING

Identify phagocytes.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

White blood cell (phagocyte).

MARK BREAKDOWN

Mark 1: White blood cell (phagocyte).

COMMON MISTAKES

- Red blood cell.
- Nerve cell.

EXAMINER TIP

Phagocytes engulf and digest pathogens.

QUESTION 30

Describe how the body prevents pathogens entering through the breathing system.

QUESTION TYPE

Topic: Infection and response | **Command word:** describe | **Marks:** 2 | **Skill tested:** Defences

WHAT THE QUESTION IS ASKING

Mention mucus, cilia and nose hairs.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Sticky mucus traps pathogens in the airways. Cilia beat to move mucus to the throat to be swallowed or coughed out. Hairs in the nose filter larger particles.

MARK BREAKDOWN

Mark 1: Mucus traps pathogens.

Mark 2: Cilia move mucus away.

COMMON MISTAKES

- Only mentioning skin.
- Saying mucus kills all pathogens.

EXAMINER TIP

First-line defences are non-specific barriers.

QUESTION 31

Explain why antibiotics do not work against viral infections.

QUESTION TYPE

Topic: Infection and response | **Command word:** explain | **Marks:** 2 | **Skill tested:** Medicines

WHAT THE QUESTION IS ASKING

Antibiotics target bacterial structures not found in viruses.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Antibiotics work by interfering with bacterial cell walls or protein synthesis. Viruses reproduce inside host cells using host machinery and have no bacterial cell wall, so antibiotics cannot target them effectively.

MARK BREAKDOWN

Mark 1: Antibiotics target bacterial structures/processes.

Mark 2: Viruses use host cells / lack bacterial targets.

COMMON MISTAKES

- Saying viruses are stronger.
- Claiming all medicines kill viruses.

EXAMINER TIP

Antibiotics are for bacteria, not viruses.

QUESTION 32

Compare active and passive immunity.

QUESTION TYPE

Topic: Infection and response | **Command word:** compare | **Marks:** 2 | **Skill tested:** Immunity

WHAT THE QUESTION IS ASKING

State how each is acquired and duration.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Active immunity is produced by the body after infection or vaccination and is often long-lasting. Passive immunity is provided by antibodies from another organism (e.g. breast milk or antiserum) and is short-term.

MARK BREAKDOWN

Mark 1: Active: body makes own antibodies.

Mark 2: Passive: receives antibodies; short-term.

COMMON MISTAKES

• Saying passive immunity is permanent.
• No comparison of duration.

EXAMINER TIP

Active = own antibodies; passive = borrowed antibodies.

QUESTION 33

Explain how vaccination helps protect individuals and communities against disease.

QUESTION TYPE

Topic: Infection and response | **Command word:** explain | **Marks:** 3 | **Skill tested:** Vaccination

WHAT THE QUESTION IS ASKING

Link memory cells and herd immunity.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Vaccines contain harmless forms of antigens that stimulate white blood cells to produce antibodies and memory cells. On real infection, a faster secondary response prevents illness. High vaccination rates reduce pathogen spread, protecting those who cannot be vaccinated.

MARK BREAKDOWN

Mark 1: Vaccine stimulates antibodies and memory cells.
Mark 2: Faster secondary response.
Mark 3: Reduced spread or herd immunity.

COMMON MISTAKES

• Saying vaccines contain live pathogens always.
• No community benefit.

EXAMINER TIP

Include both individual and population protection.

QUESTION 34

Describe how monoclonal antibodies can be used in pregnancy tests.

QUESTION TYPE

Topic: Infection and response | **Command word:** describe | **Marks:** 3 | **Skill tested:** Biotechnology

WHAT THE QUESTION IS ASKING

Explain antigen–antibody binding on test strip.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Monoclonal antibodies on the test strip bind to hCG hormone in urine. When hCG is present, coloured beads bound to mobile antibodies accumulate at the test line, producing a visible coloured band.

MARK BREAKDOWN

Mark 1: hCG detected by antibodies.

Mark 2: Binding at test line.

Mark 3: Visible colour change.

COMMON MISTAKES

- Saying antibodies are produced in the test.
- No mention of hCG.

EXAMINER TIP

Monoclonal antibodies bind to one specific antigen.

QUESTION 35

A new strain of bacterium is resistant to penicillin. Explain how natural selection has led to this.

QUESTION TYPE

Topic: Infection and response | **Command word:** explain | **Marks:** 4 | **Skill tested:** Antibiotic resistance

WHAT THE QUESTION IS ASKING

Variation, selection pressure, reproduction of resistant bacteria.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Random mutations produce some bacteria resistant to penicillin. When penicillin is used, susceptible bacteria die but resistant ones survive and reproduce. Over time the proportion of resistant bacteria increases in the population.

MARK BREAKDOWN

Mark 1: Variation or mutation gives resistance.

Mark 2: Antibiotic selects survivors.

Mark 3: Resistant bacteria reproduce.

Mark 4: Increase in resistant proportion.

COMMON MISTAKES

- Saying bacteria evolve to need antibiotics.
- Lamarckian 'trying to survive'.

EXAMINER TIP

Selection acts on existing variation — bacteria do not 'try' to become resistant.

QUESTION 36

Evaluate the use of antibiotics to treat mild throat infections.

QUESTION TYPE

Topic: Infection and response | **Command word:** evaluate | **Marks:** 4 | **Skill tested:** Medicine

WHAT THE QUESTION IS ASKING

Weigh symptom relief against resistance and side effects.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Antibiotics may shorten bacterial throat infections but many sore throats are viral and will not respond. Overuse selects resistant bacteria and can cause side effects. Rest and fluids may be sufficient for mild cases. Reserve antibiotics for confirmed bacterial infection.

MARK BREAKDOWN

Mark 1: May help if bacterial.

Mark 2: Ineffective if viral.

Mark 3: Overuse drives resistance.

Mark 4: Balanced judgement on mild cases.

COMMON MISTAKES

- Always using antibiotics.
- Ignoring resistance.

EXAMINER TIP

Evaluation needs advantage, limitation and conclusion.

QUESTION 37

Describe how you would prepare and observe a bacterial culture on agar using aseptic technique. Include safety and how to reduce contamination.

QUESTION TYPE

Topic: Infection and response | **Command word:** describe | **Marks:** 6 | **Skill tested:** Microbiology practical

WHAT THE QUESTION IS ASKING

Full aseptic technique method with evaluation.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Sterilise inoculating loop by flaming until red hot and cooling. Lift lid of Petri dish slightly and streak sample on agar. Seal dish with tape (not airtight). Incubate upside down at 25 °C in school labs. Work near a Bunsen flame to create updraft. Disinfect bench before and after. Do not open plates after incubation; seal and dispose safely.

MARK BREAKDOWN

Mark 1: Loop sterilised by flaming.

Mark 2: Minimal lid opening.

Mark 3: Correct incubation temperature for school.

Mark 4: Flame or disinfection used.

Mark 5: Sealed disposal.

Mark 6: Logical safe sequence.

COMMON MISTAKES

- Incubating at 37 °C in open plates.
- No sterilisation of loop.

EXAMINER TIP

School labs use 25 °C to reduce pathogen risk.

GRADE 7–9 EXTENSION

Link aseptic technique to preventing unwanted microbes.

QUESTION 38

The area of a Petri dish colony doubles every 3 hours. After 3 hours the colony area is 12 mm^2. Calculate the area after 9 hours if growth continues at the same rate.

QUESTION TYPE

Topic: Infection and response | **Command word:** calculate | **Marks:** 3 | **Skill tested:** Calculation

WHAT THE QUESTION IS ASKING

Three doubling periods occur in 9 hours.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

9 hours = three doubling periods. Area = $12 \times 2 \times 2 \times 2 = 12 \times 8 = 96$ mm^2.

MARK BREAKDOWN

Mark 1: Recognises three doublings.

Mark 2: Multiplies by 2 three times.

Mark 3: 96 mm^2.

COMMON MISTAKES

- Adding instead of multiplying.
- Only one doubling.

EXAMINER TIP

Exponential growth multiplies; do not add the increase each time.

QUESTION 39

Describe how you would investigate the effect of an antibiotic on bacterial growth using pre-poured agar plates.

QUESTION TYPE

Topic: Infection and response | **Command word:** describe | **Marks:** 4 | **Skill tested:** Required practical — antimicrobial discs

WHAT THE QUESTION IS ASKING

Method for antibiotic disc diffusion test.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Use aseptic technique to streak bacteria evenly on agar. Place antibiotic discs on the surface with sterile forceps. Incubate at 25 °C for 48 hours. Measure the clear zone (zone of inhibition) around each disc. Larger zones indicate greater effectiveness.

MARK BREAKDOWN

Mark 1: Even bacterial lawn.

Mark 2: Antibiotic discs placed aseptically.

Mark 3: Zone of inhibition measured.

Mark 4: Link zone size to effectiveness.

COMMON MISTAKES

- Measuring colony count only.
- Opening plates during incubation.

EXAMINER TIP

Clear zone = bacteria killed by antibiotic diffusing from disc.

Section 2: Bioenergetics

QUESTION 40

State the word equation for aerobic respiration.

QUESTION TYPE

Topic: Bioenergetics | **Command word:** state | **Marks:** 1 | **Skill tested:** Respiration

WHAT THE QUESTION IS ASKING

Recall reactants and products.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Glucose + oxygen $\rightarrow$ carbon dioxide + water

MARK BREAKDOWN

Mark 1: Glucose + oxygen $\rightarrow$ carbon dioxide + water

COMMON MISTAKES

- Including light.
- Lactic acid as product.

EXAMINER TIP

Aerobic respiration needs oxygen and releases CO■.

QUESTION 41

Give the name of the green pigment that absorbs light in photosynthesis.

QUESTION TYPE

Topic: Bioenergetics | **Command word:** give | **Marks:** 1 | **Skill tested:** Photosynthesis

WHAT THE QUESTION IS ASKING

Name the pigment in chloroplasts.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Chlorophyll.

MARK BREAKDOWN

Mark 1: Chlorophyll.

COMMON MISTAKES

- Carotene only without chlorophyll.
- Haemoglobin.

EXAMINER TIP

Chlorophyll is in chloroplasts.

QUESTION 42

State where aerobic respiration occurs in a cell.

QUESTION TYPE

Topic: Bioenergetics | **Command word:** state | **Marks:** 1 | **Skill tested:** Respiration

WHAT THE QUESTION IS ASKING

Identify the organelle.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Mitochondria.

MARK BREAKDOWN

Mark 1: Mitochondria.

COMMON MISTAKES

- Chloroplasts.
- Nucleus.

EXAMINER TIP

Mitochondria are the site of aerobic respiration.

QUESTION 43

Describe the effect of increasing light intensity on the rate of photosynthesis at low light levels.

QUESTION TYPE

Topic: Bioenergetics | **Command word:** describe | **Marks:** 2 | **Skill tested:** Limiting factors

WHAT THE QUESTION IS ASKING

Rate increases then may plateau.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

As light intensity increases, the rate of photosynthesis increases because more light energy is available for the light-dependent reactions. At low light, light intensity is the limiting factor.

MARK BREAKDOWN

Mark 1: Rate increases with light.

Mark 2: Light is limiting factor at low intensity.

COMMON MISTAKES

- Saying rate always increases without limit.
- Confusing with respiration only.

EXAMINER TIP

Limiting factor = factor restricting rate.

QUESTION 44

Explain why plants respire all the time.

QUESTION TYPE

Topic: Bioenergetics | **Command word:** explain | **Marks:** 3 | **Skill tested:** Respiration

WHAT THE QUESTION IS ASKING

ATP needed for life processes day and night.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Respiration releases ATP needed for active transport, protein synthesis, cell division and movement. These processes continue day and night, so respiration must continue even when photosynthesis stops in the dark.

MARK BREAKDOWN

Mark 1: ATP needed for life processes.

Mark 2: Named process requiring ATP.

Mark 3: Continues in dark / not only in light.

COMMON MISTAKES

- Saying respiration only happens at night.
- Confusing with breathing.

EXAMINER TIP

Respiration ≠ breathing; it happens in all living cells.

QUESTION 45

Compare aerobic and anaerobic respiration in yeast.

QUESTION TYPE

Topic: Bioenergetics | **Command word:** compare | **Marks:** 2 | **Skill tested:** Respiration

WHAT THE QUESTION IS ASKING

Products and energy yield.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Both release energy. Aerobic respiration in yeast produces carbon dioxide and water with a high ATP yield. Anaerobic respiration produces ethanol and carbon dioxide with a lower ATP yield.

MARK BREAKDOWN

Mark 1: Both release energy.

Mark 2: Different products or ATP yield stated.

COMMON MISTAKES

- Saying anaerobic produces lactic acid in yeast.
- Equal ATP yield.

EXAMINER TIP

Yeast anaerobic = ethanol fermentation.

QUESTION 46

Explain how the rate of photosynthesis is limited by carbon dioxide concentration.

QUESTION TYPE

Topic: Bioenergetics | **Command word:** explain | **Marks:** 3 | **Skill tested:** Limiting factors

WHAT THE QUESTION IS ASKING

CO■ needed for Calvin cycle; plateau when other factor limits.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Carbon dioxide is a reactant in photosynthesis. Increasing CO■ increases the rate until another factor such as light or temperature becomes limiting, after which the rate plateaus.

MARK BREAKDOWN

Mark 1: CO■ is a reactant.

Mark 2: Rate increases then plateaus.

Mark 3: Another factor becomes limiting.

COMMON MISTAKES

- Saying CO■ is a product.
- No plateau mentioned.

EXAMINER TIP

Only one factor limits at a time at steady state.

QUESTION 47

Describe an experiment to investigate the effect of temperature on the rate of photosynthesis using pondweed.

QUESTION TYPE

Topic: Bioenergetics | **Command word:** describe | **Marks:** 2 | **Skill tested:** Investigations

WHAT THE QUESTION IS ASKING

Count bubbles at different temperatures with controlled light and CO■.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Place pondweed in water with sodium hydrogencarbonate (CO■ source) under a lamp at fixed distance. Count bubbles per minute at different water temperatures in a water bath. Keep light intensity and CO■ concentration constant.

MARK BREAKDOWN

Mark 1: Bubble counting as rate measure.

Mark 2: Temperature changed with light and CO■ controlled.

COMMON MISTAKES

- Changing light and temperature together.
- No CO■ source.

EXAMINER TIP

Count bubbles for the same time at each temperature.

QUESTION 48

A farmer grows tomatoes in a greenhouse. Explain how adjusting light, CO■ and temperature can increase crop yield.

QUESTION TYPE

Topic: Bioenergetics | **Command word:** explain | **Marks:** 4 | **Skill tested:** Applications

WHAT THE QUESTION IS ASKING

Optimise photosynthesis without excessive respiration cost.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Extra lighting extends photosynthesis time. Enriched CO■ increases carbon fixation when light is sufficient. Optimum warmth increases enzyme activity in photosynthesis. Together they raise sugar production for growth, but excessive heat may increase respiration losses.

MARK BREAKDOWN

Mark 1: Light increases photosynthesis time/rate.

Mark 2: CO■ increases rate.

Mark 3: Temperature affects enzymes.

Mark 4: Link to yield or growth.

COMMON MISTAKES

- Only mentioning one factor.
- Ignoring respiration at high temperature.

EXAMINER TIP

Commercial growers balance photosynthesis gain against cost.

QUESTION 49

Explain why anaerobic respiration in muscles during vigorous exercise causes fatigue.

QUESTION TYPE

Topic: Bioenergetics | **Command word:** explain | **Marks:** 4 | **Skill tested:** Exercise

WHAT THE QUESTION IS ASKING

Lactic acid and oxygen debt.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

When oxygen supply is insufficient, muscles respire anaerobically producing lactic acid. Lactic acid accumulation causes muscle fatigue and oxygen debt must be repaid after exercise to oxidise it.

MARK BREAKDOWN

Mark 1: Anaerobic in muscles.

Mark 2: Lactic acid produced.

Mark 3: Fatigue or oxygen debt.

Mark 4: Recovery need.

COMMON MISTAKES

- Ethanol in muscles.
- No oxygen debt.

EXAMINER TIP

Muscle anaerobic → lactic acid, not ethanol.

QUESTION 50

Describe how you would investigate the effect of light colour on the rate of photosynthesis using Elodea and a lamp with coloured filters. Include fair test details and expected results.

QUESTION TYPE

Topic: Bioenergetics | **Command word:** describe | **Marks:** 6 | **Skill tested:** Photosynthesis practical

WHAT THE QUESTION IS ASKING

Full investigation with variables and prediction.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Place Elodea in sodium hydrogencarbonate solution at fixed temperature. Position lamp at constant distance with red, green and blue filters in turn. Count bubbles per minute for each colour, repeating and calculating mean. Keep plant piece size and volume of water constant. Expect highest rate under red/blue as chlorophyll absorbs these best; green lowest as it is reflected.

MARK BREAKDOWN

Mark 1: Same distance and temperature.

Mark 2: Filters change light colour only.

Mark 3: Bubble rate measured with repeats.

Mark 4: Prediction linked to chlorophyll absorption.

Mark 5: Fair test variables listed.

Mark 6: Logical sequence.

COMMON MISTAKES

- Changing lamp distance for each colour.
- No repeats.

EXAMINER TIP

Chlorophyll absorbs red and blue most; reflects green.

GRADE 7–9 EXTENSION

Relate results to absorption spectrum of chlorophyll.

QUESTION 51

A plant produces 0.48 g of biomass in 2 hours. Calculate the rate of biomass production in g/h.

QUESTION TYPE

Topic: Bioenergetics | **Command word:** calculate | **Marks:** 3 | **Skill tested:** Calculation

WHAT THE QUESTION IS ASKING

Rate = amount ÷ time.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Rate = 0.48 ÷ 2 = 0.24 g/h.

MARK BREAKDOWN

Mark 1: Correct formula rate = amount ÷ time.

Mark 2: Correct division.

Mark 3: 0.24 g/h with unit.

COMMON MISTAKES

- Multiplying by 2.
- Missing unit.

EXAMINER TIP

Rate always includes 'per unit time'.

QUESTION 52

Describe how you would investigate the effect of sugar concentration on yeast respiration by measuring carbon dioxide production.

QUESTION TYPE

Topic: Bioenergetics | **Command word:** describe | **Marks:** 4 | **Skill tested:** Required practical — yeast

WHAT THE QUESTION IS ASKING

Use yeast in sugar solutions with gas collection.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Mix equal masses of yeast with different sugar concentrations in warm water. Attach boiling tubes to gas syringes or inverted measuring cylinders in a water bath at 35 °C. Measure volume of CO_2 collected in five minutes. Keep yeast mass, volume and temperature constant.

MARK BREAKDOWN

Mark 1: Different sugar concentrations.

Mark 2: CO_2 volume measured.

Mark 3: Temperature and yeast mass controlled.

Mark 4: Same time interval.

COMMON MISTAKES

- Measuring temperature change only.
- No control of yeast amount.

EXAMINER TIP

Warm water activates yeast; too hot kills it.

Section 2: Homeostasis and response

QUESTION 53

State the name of the gland that produces insulin.

QUESTION TYPE

Topic: Homeostasis and response | **Command word:** state | **Marks:** 1 | **Skill tested:** Hormones

WHAT THE QUESTION IS ASKING

Recall endocrine source of insulin.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Pancreas.

MARK BREAKDOWN

Mark 1: Pancreas.

COMMON MISTAKES

- Pituitary.
- Adrenal gland.

EXAMINER TIP

Insulin is produced by β cells in the pancreas.

QUESTION 54

Give the name of the type of neurone that carries impulses from receptors to the central nervous system.

QUESTION TYPE

Topic: Homeostasis and response | **Command word:** give | **Marks:** 1 | **Skill tested:** Nervous system

WHAT THE QUESTION IS ASKING

Name sensory neurones.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Sensory neurone.

MARK BREAKDOWN

Mark 1: Sensory neurone.

COMMON MISTAKES

- Motor neurone.
- Relay neurone only.

EXAMINER TIP

Sensory → CNS → motor pathway.

QUESTION 55

State what is meant by homeostasis.

QUESTION TYPE

Topic: Homeostasis and response | **Command word:** state | **Marks:** 1 | **Skill tested:** Homeostasis

WHAT THE QUESTION IS ASKING

Define maintenance of internal conditions.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Homeostasis is the regulation of internal conditions to maintain optimum conditions for cell function.

MARK BREAKDOWN

Mark 1: Homeostasis is the regulation of internal conditions to maintain optimum conditions for cell function.

COMMON MISTAKES

- Keeping the body warm only.
- Controlling movement.

EXAMINER TIP

Homeostasis covers water, temperature, blood glucose and more.

QUESTION 56

Describe the function of the skin in thermoregulation when body temperature rises.

QUESTION TYPE

Topic: Homeostasis and response | **Command word:** describe | **Marks:** 2 | **Skill tested:** Temperature control

WHAT THE QUESTION IS ASKING

Vasodilation and sweating.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Blood vessels near the skin surface dilate (vasodilation) so more blood flows near the surface, transferring heat to the environment. Sweat glands release sweat which evaporates, removing heat energy.

MARK BREAKDOWN

Mark 1: Vasodilation or more blood near surface.

Mark 2: Sweating and evaporation removes heat.

COMMON MISTAKES

- Vasoconstriction when hot.

• Saying hair stands up for cooling.

EXAMINER TIP

Hot → vasodilation + sweat; cold → vasoconstriction + shivering.

QUESTION 57

Explain why type 1 diabetes patients need insulin injections.

QUESTION TYPE

Topic: Homeostasis and response | **Command word:** explain | **Marks:** 2 | **Skill tested:** Diabetes

WHAT THE QUESTION IS ASKING

No insulin production leads to high blood glucose.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

In type 1 diabetes the pancreas produces little or no insulin, so glucose is not taken up by cells effectively and blood glucose stays dangerously high. Insulin injections lower blood glucose to a safe level.

MARK BREAKDOWN

Mark 1: Little or no insulin produced.

Mark 2: Blood glucose stays high without treatment.

Mark 3: Insulin lowers blood glucose.

COMMON MISTAKES

• Confusing type 1 and type 2.

• Saying insulin raises glucose.

EXAMINER TIP

Type 1 = no insulin; type 2 = cells resist insulin.

QUESTION 58

Compare the nervous and hormonal systems.

QUESTION TYPE

Topic: Homeostasis and response | **Command word:** compare | **Marks:** 2 | **Skill tested:** Coordination

WHAT THE QUESTION IS ASKING

Speed, duration and pathway differences.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The nervous system uses electrical impulses along neurones for fast, short-lived, targeted responses. The hormonal system uses hormones in the blood for slower, longer-lasting, widespread responses.

MARK BREAKDOWN

Mark 1: Nervous: fast/electrical.

Mark 2: Hormonal: slower/chemical in blood.

COMMON MISTAKES

- Saying hormones are faster.
- No duration difference.

EXAMINER TIP

Nervous = quick and local; hormones = slow and widespread.

QUESTION 59

Explain how a reflex arc protects the body from harm.

QUESTION TYPE

Topic: Homeostasis and response | **Command word:** explain | **Marks:** 3 | **Skill tested:** Reflexes

WHAT THE QUESTION IS ASKING

Receptor to effector via spinal cord without conscious brain.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

A stimulus is detected by receptors and an impulse passes along a sensory neurone to the spinal cord. A relay neurone connects to a motor neurone, which stimulates an effector (muscle or gland) to remove the body part from danger before the brain processes the sensation.

MARK BREAKDOWN

Mark 1: Receptor detects stimulus.

Mark 2: Spinal cord relay.

Mark 3: Motor neurone to effector for rapid response.

COMMON MISTAKES

- Impulse goes to brain first.
- No effector mentioned.

EXAMINER TIP

Reflexes are automatic and faster than conscious responses.

QUESTION 60

Describe how the kidneys respond when blood water potential is too low.

QUESTION TYPE

Topic: Homeostasis and response | **Command word:** describe | **Marks:** 3 | **Skill tested:** Water balance

WHAT THE QUESTION IS ASKING

More ADH, more water reabsorption.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.

5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The brain detects low water potential and releases more ADH. ADH makes kidney tubules more permeable so more water is reabsorbed into the blood, producing a smaller volume of concentrated urine.

MARK BREAKDOWN

Mark 1: ADH increases.

Mark 2: More water reabsorbed in kidneys.

Mark 3: Concentrated urine produced.

COMMON MISTAKES

- Less reabsorption when dehydrated.
- ADH from kidneys.

EXAMINER TIP

Low water → more ADH → save water.

QUESTION 61

A student eats a large sugary meal. Explain how the body returns blood glucose to normal.

QUESTION TYPE

Topic: Homeostasis and response | **Command word:** explain | **Marks:** 4 | **Skill tested:** Blood glucose control

WHAT THE QUESTION IS ASKING

Insulin from pancreas, uptake and storage as glycogen.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Blood glucose rises after the meal. The pancreas secretes insulin, which causes liver and muscle cells to take up glucose and convert it to glycogen for storage. Blood glucose falls back to normal. If it drops too low, glucagon is released.

MARK BREAKDOWN

Mark 1: Glucose rises after meal.

Mark 2: Insulin secreted.

Mark 3: Glucose taken up or stored as glycogen.

Mark 4: Returns to normal.

COMMON MISTAKES

- Only glucagon after eating.
- Insulin raises glucose.

EXAMINER TIP

After eating → insulin; when low → glucagon.

QUESTION 62

Evaluate the use of plant hormones such as auxin in agriculture.

QUESTION TYPE

Topic: Homeostasis and response | **Command word:** evaluate | **Marks:** 4 | **Skill tested:** Plant hormones

WHAT THE QUESTION IS ASKING

Weigh benefits in rooting and weed control against cost and ethics.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Auxin rooting powder helps cuttings develop roots quickly, saving time. Selective weedkillers based on auxin mimicry kill broad-leaved weeds in grass crops. However, overuse may affect non-target plants and consumers may prefer reduced chemical use. Overall useful when applied carefully.

MARK BREAKDOWN

Mark 1: Benefit: rooting or weed control.

Mark 2: Selective action explained.

Mark 3: Limitation or environmental concern.

Mark 4: Supported conclusion.

COMMON MISTAKES

• Only advantages.

• Saying auxin kills all plants equally.

EXAMINER TIP

Selective weedkillers exploit different auxin sensitivity.

QUESTION 63

Describe an investigation to show phototropism in plant seedlings. Include how you would measure the response and control variables.

QUESTION TYPE

Topic: Homeostasis and response | **Command word:** describe | **Marks:** 6 | **Skill tested:** Plant responses practical

WHAT THE QUESTION IS ASKING

Compare light from one side with controls.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Grow identical seedlings in moist cotton wool. Place one group with unidirectional light from one side, another in a cardboard box with a single hole, and a control under uniform light. After several days, measure angle of shoot curvature from vertical. Keep temperature and water constant. Repeat with ten seedlings per group and calculate mean angle.

MARK BREAKDOWN

Mark 1: Unidirectional light setup.
Mark 2: Control under uniform light.
Mark 3: Angle or curvature measured.
Mark 4: Variables controlled.
Mark 5: Repeats and mean.
Mark 6: Clear time period.

COMMON MISTAKES

- No control group.
- Measuring root length only.

EXAMINER TIP

Auxin redistributes to shaded side causing unequal growth.

GRADE 7–9 EXTENSION

Link curvature to auxin and differential elongation.

QUESTION 64

A person has body mass 72 kg and height 1.80 m. Calculate their BMI using BMI = mass (kg) ÷ [height (m)]^2.

QUESTION TYPE

Topic: Homeostasis and response | **Command word:** calculate | **Marks:** 3 | **Skill tested:** Calculation

WHAT THE QUESTION IS ASKING

Square height before dividing mass.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Height^2 = 1.80^2 = 3.24 m^2. BMI = 72 ÷ 3.24 = 22.2 kg/m^2 (3 s.f.).

MARK BREAKDOWN

Mark 1: Height squared correctly.
Mark 2: Division set up.
Mark 3: BMI ≈ 22.2 with unit.

COMMON MISTAKES

- Not squaring height.
- Using cm without converting.

EXAMINER TIP

Always convert height to metres before squaring.

QUESTION 65

Describe how you would investigate the effect of caffeine on reaction time using a ruler-drop test.

QUESTION TYPE

Topic: Homeostasis and response | **Command word:** describe | **Marks:** 4 | **Skill tested:** Required practical — reaction time

WHAT THE QUESTION IS ASKING

Fair test comparing reaction times with and without caffeine.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Measure baseline reaction time: partner drops ruler, subject catches and record distance fallen. Convert to time. Repeat after controlled caffeine dose at same time of day, with same sleep. Use at least ten catches per condition and calculate mean. Exclude anomalous drops. Keep same hand and ruler width.

MARK BREAKDOWN

Mark 1: Ruler-drop method described.

Mark 2: Before/after caffeine comparison.

Mark 3: Repeats and mean.

Mark 4: Control variables named.

COMMON MISTAKES

- No repeats.
- Changing multiple variables at once.

EXAMINER TIP

Lower distance = faster reaction; convert using $d = \frac{1}{2}gt^2$ if needed.

Section 2: Inheritance, variation and evolution

QUESTION 66

State the name of the molecule that carries genetic information in the nucleus.

QUESTION TYPE

Topic: Inheritance, variation and evolution | **Command word:** state | **Marks:** 1 | **Skill tested:** Genetics

WHAT THE QUESTION IS ASKING

Identify genetic material.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

DNA (deoxyribonucleic acid).

MARK BREAKDOWN

Mark 1: DNA (deoxyribonucleic acid).

COMMON MISTAKES

- Protein.
- Glucose.

EXAMINER TIP

DNA is the genetic material in chromosomes.

QUESTION 67

Give the term for different versions of the same gene.

QUESTION TYPE

Topic: Inheritance, variation and evolution | **Command word:** give | **Marks:** 1 | **Skill tested:** Genetics

WHAT THE QUESTION IS ASKING

Name alleles.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Alleles.

MARK BREAKDOWN

Mark 1: Alleles.

COMMON MISTAKES

- Chromosomes.
- Nucleotides only.

EXAMINER TIP

Alleles are versions of a gene.

QUESTION 68

State what is meant by a dominant allele.

QUESTION TYPE

Topic: Inheritance, variation and evolution | **Command word:** state | **Marks:** 1 | **Skill tested:** Inheritance

WHAT THE QUESTION IS ASKING

Define dominance.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

An allele that is always expressed in the phenotype when present.

MARK BREAKDOWN

Mark 1: An allele that is always expressed in the phenotype when present.

COMMON MISTAKES

- The most common allele.
- An allele on the X chromosome only.

EXAMINER TIP

Dominant shows with one copy; recessive needs two.

QUESTION 69

Describe the structure of DNA including the complementary base pairs.

QUESTION TYPE

Topic: Inheritance, variation and evolution | **Command word:** describe | **Marks:** 2 | **Skill tested:** DNA structure

WHAT THE QUESTION IS ASKING

Double helix, bases A-T and C-G.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

DNA is a double helix made of two strands. Bases pair specifically: adenine with thymine and cytosine with guanine, held by hydrogen bonds.

MARK BREAKDOWN

Mark 1: Double helix or two strands.

Mark 2: A-T and C-G pairing.

COMMON MISTAKES

- A pairs with G.
- Single strand only.

EXAMINER TIP

Chargaff's rules: A=T, C=G.

QUESTION 70

Explain why siblings inherit different combinations of alleles from the same parents.

QUESTION TYPE

Topic: Inheritance, variation and evolution | **Command word:** explain | **Marks:** 3 | **Skill tested:** Meiosis

WHAT THE QUESTION IS ASKING

Independent assortment and fertilisation randomness.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

During meiosis, gametes receive one chromosome from each homologous pair at random. Fertilisation combines gametes randomly, so each offspring gets a unique allele combination unless identical twins.

MARK BREAKDOWN

Mark 1: Gametes genetically different due to meiosis.

Mark 2: Random fertilisation.

Mark 3: Unique combination in offspring.

COMMON MISTAKES

- Mitosis in gametes.
- Identical gametes from parents.

EXAMINER TIP

Meiosis produces variation; fertilisation shuffles it.

QUESTION 71

Compare genetic and environmental variation.

QUESTION TYPE

Topic: Inheritance, variation and evolution | **Command word:** compare | **Marks:** 2 | **Skill tested:** Variation

WHAT THE QUESTION IS ASKING

Origin and examples.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Genetic variation is inherited from parents via alleles, e.g. blood group. Environmental variation is caused by lifestyle or surroundings, e.g. language spoken or scars. Most characteristics are influenced by both.

MARK BREAKDOWN

Mark 1: Genetic: inherited alleles.

Mark 2: Environmental: caused by surroundings.

COMMON MISTAKES

- Saying all variation is genetic.
- No example.

EXAMINER TIP

Give one example of each type.

QUESTION 72

Explain how natural selection can lead to evolution of antibiotic resistance in bacteria.

QUESTION TYPE

Topic: Inheritance, variation and evolution | **Command word:** explain | **Marks:** 3 | **Skill tested:** Evolution

WHAT THE QUESTION IS ASKING

Variation, selection, inheritance over generations.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Random mutations create resistant bacteria. Antibiotics kill non-resistant forms, leaving resistant ones to reproduce. Alleles for resistance become more common over generations, changing the population — evolution.

MARK BREAKDOWN

Mark 1: Mutation gives variation.

Mark 2: Selection by antibiotic.

Mark 3: Survivors reproduce; allele frequency changes.

COMMON MISTAKES

- Bacteria change to resist.
- All bacteria mutate at once.

EXAMINER TIP

Evolution = change in allele frequency in a population.

QUESTION 73

Describe how to interpret a pedigree diagram showing inheritance of a recessive disorder.

QUESTION TYPE

Topic: Inheritance, variation and evolution | **Command word:** describe | **Marks:** 2 | **Skill tested:** Pedigree analysis

WHAT THE QUESTION IS ASKING

Identify carriers and affected individuals.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Affected individuals have two recessive alleles and may skip generations if parents are carriers. Carriers are unaffected heterozygotes. If two carriers have children, about one quarter may be affected in each pregnancy.

MARK BREAKDOWN

Mark 1: Recessive can skip generations.

Mark 2: Carriers heterozygous unaffected or offspring ratio stated.

COMMON MISTAKES

- Dominant pattern described.
- All children of carriers affected.

EXAMINER TIP

Recessive disorders often appear in siblings with unaffected parents.

QUESTION 74

In pea plants, allele T for tall is dominant to t for short. Cross two heterozygous tall plants. Explain the expected phenotype ratio in offspring.

QUESTION TYPE

Topic: Inheritance, variation and evolution | **Command word:** explain | **Marks:** 4 | **Skill tested:** Genetic crosses

WHAT THE QUESTION IS ASKING

Punnett square Tt × Tt gives 3:1.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Parents are Tt × Tt. Gametes T and t from each. Offspring genotypes TT, Tt, Tt, tt — phenotypically three tall to one short because T is dominant.

MARK BREAKDOWN

Mark 1: Parents Tt identified.

Mark 2: Gametes T and t.

Mark 3: Ratio 3 tall : 1 short.

Mark 4: tt is short phenotype.

COMMON MISTAKES

- 1:1 ratio.
- tt tall phenotype.

EXAMINER TIP

Draw Punnett square for four genotype outcomes.

QUESTION 75

Explain how the fossil record provides evidence for evolution.

QUESTION TYPE

Topic: Inheritance, variation and evolution | **Command word:** explain | **Marks:** 4 | **Skill tested:** Evidence for evolution

WHAT THE QUESTION IS ASKING

Gradual change, transitional forms, extinction.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Fossils show organisms that lived in the past and how species change over geological time. Transitional fossils link groups, and some species in fossils no longer exist, supporting change and extinction over millions of years.

MARK BREAKDOWN

Mark 1: Fossils show past life.

Mark 2: Change over time.

Mark 3: Transitional or extinct forms.

Mark 4: Link to evolution.

COMMON MISTAKES

- All fossils identical to today.
- Fossils prove creation in one day.

EXAMINER TIP

Older rocks generally contain simpler or different forms.

QUESTION 76

Describe how selective breeding is used to improve crop plants. In your answer, explain steps taken, benefits and one ethical or biological limitation.

QUESTION TYPE

Topic: Inheritance, variation and evolution | **Command word:** describe | **Marks:** 6 | **Skill tested:** Selective breeding

WHAT THE QUESTION IS ASKING

Choose parents, breed, repeat over generations.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Farmers select plants with desired traits such as high yield or disease resistance and breed them together. Offspring showing the trait are selected and bred again over many generations to increase allele frequency. Benefits include higher food production without GM. Limitations: reduced gene pool may increase disease vulnerability or it takes many generations.

MARK BREAKDOWN

Mark 1: Selection of desired trait.
Mark 2: Breed selected offspring.
Mark 3: Repeat over generations.
Mark 4: Benefit stated.
Mark 5: Limitation stated.
Mark 6: Coherent account.

COMMON MISTAKES

- One generation only.
- Same as genetic modification.

EXAMINER TIP

Selective breeding uses natural variation; GM inserts genes directly.

GRADE 7–9 EXTENSION

Compare with cloning and GM for top grades.

QUESTION 77

In a population of 200 plants, 18 show a recessive phenotype. Calculate the percentage of plants homozygous recessive.

QUESTION TYPE

Topic: Inheritance, variation and evolution | **Command word:** calculate | **Marks:** 3 | **Skill tested:** Calculation

WHAT THE QUESTION IS ASKING

Percentage = (number ÷ total) × 100.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Percentage = (18 ÷ 200) × 100 = 9%.

MARK BREAKDOWN

Mark 1: Correct formula.
Mark 2: Correct fraction 18/200.
Mark 3: 9%.

COMMON MISTAKES

- Using 18/100.
- Forgetting to multiply by 100.

EXAMINER TIP

Percentage must relate part to whole population.

QUESTION 78

Describe how you would extract DNA from fruit such as strawberries using classroom reagents.

QUESTION TYPE

Topic: Inheritance, variation and evolution | **Command word:** describe | **Marks:** 4 | **Skill tested:** Required practical — DNA extraction

WHAT THE QUESTION IS ASKING

Crush, detergent, heat, cold ethanol precipitation.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Crush fruit with extraction buffer containing detergent and salt to break membranes and release DNA. Filter to remove solids. Add ice-cold ethanol to filtrate; white strands precipitate at the interface. Spool DNA on a glass rod.

MARK BREAKDOWN

Mark 1: Mechanical breakdown of cells.

Mark 2: Detergent breaks membranes.

Mark 3: Cold ethanol precipitates DNA.

Mark 4: Observable white strands.

COMMON MISTAKES

- Using hot ethanol.
- No filtration step.

EXAMINER TIP

Cold ethanol makes DNA insoluble so it visible.

Section 2: Ecology

QUESTION 79

State what is meant by a community in ecology.

QUESTION TYPE

Topic: Ecology | **Command word:** state | **Marks:** 1 | **Skill tested:** Ecosystems

WHAT THE QUESTION IS ASKING

Define community.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

All the populations of different species living in the same habitat.

MARK BREAKDOWN

Mark 1: All the populations of different species living in the same habitat.

COMMON MISTAKES

- All organisms in the world.
- One species only.

EXAMINER TIP

Community = all species in a habitat.

QUESTION 80

Give one abiotic factor that affects ecosystems.

QUESTION TYPE

Topic: Ecology | **Command word:** give | **Marks:** 1 | **Skill tested:** Abiotic factors

WHAT THE QUESTION IS ASKING

Name a non-living factor.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Light intensity (or temperature, water availability, soil pH, wind).

MARK BREAKDOWN

Mark 1: Light intensity (or temperature, water availability, soil pH, wind).

COMMON MISTAKES

- Predation.
- Competition between animals only.

EXAMINER TIP

Abiotic = non-living physical factor.

QUESTION 81

Name the process that converts atmospheric nitrogen into usable compounds by bacteria.

QUESTION TYPE

Topic: Ecology | **Command word:** state | **Marks:** 1 | **Skill tested:** Cycles

WHAT THE QUESTION IS ASKING

Nitrogen fixation.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Nitrogen fixation.

MARK BREAKDOWN

Mark 1: Nitrogen fixation.

COMMON MISTAKES

- Denitrification.
- Photosynthesis.

EXAMINER TIP

Rhizobia in root nodules fix nitrogen.

QUESTION 82

Describe the role of decomposers in the carbon cycle.

QUESTION TYPE

Topic: Ecology | **Command word:** describe | **Marks:** 2 | **Skill tested:** Carbon cycle

WHAT THE QUESTION IS ASKING

Break down dead material releasing CO■.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Decomposers such as bacteria and fungi break down dead organisms and waste, respiring and returning carbon dioxide to the atmosphere.

MARK BREAKDOWN

Mark 1: Break down dead material.

Mark 2: Release CO■ by respiration.

COMMON MISTAKES

- Saying decomposers photosynthesise.
- Locking carbon forever.

EXAMINER TIP

Decomposers complete the cycle back to CO■.

QUESTION 83

Explain why bioaccumulation of pesticides is a problem for top predators.

QUESTION TYPE

Topic: Ecology | **Command word:** explain | **Marks:** 3 | **Skill tested:** Pollution

WHAT THE QUESTION IS ASKING

Toxins not excreted accumulate up food chain.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Pesticides are not easily broken down and accumulate in fat. Each trophic level eats many organisms from the level below, so concentration increases towards top predators, causing population decline.

MARK BREAKDOWN

Mark 1: Persistent chemicals accumulate.

Mark 2: Increase up food chain.

Mark 3: Harm to top predators.

COMMON MISTAKES

- Toxins dilute at top.
- Only affects plants.

EXAMINER TIP

Biomagnification = increasing concentration up levels.

QUESTION 84

Compare intraspecific and interspecific competition.

QUESTION TYPE

Topic: Ecology | **Command word:** compare | **Marks:** 2 | **Skill tested:** Competition

WHAT THE QUESTION IS ASKING

Same vs different species.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Intraspecific competition is between individuals of the same species for mates, food or territory. Interspecific competition is between different species for shared resources such as food or space.

MARK BREAKDOWN

Mark 1: Intraspecific: same species.

Mark 2: Interspecific: different species.

COMMON MISTAKES

- Same definition for both.
- No resource mentioned.

EXAMINER TIP

Intra = within species; inter = between species.

QUESTION 85

Explain how deforestation can contribute to climate change.

QUESTION TYPE

Topic: Ecology | **Command word:** explain | **Marks:** 3 | **Skill tested:** Human impact

WHAT THE QUESTION IS ASKING

Less CO■ uptake, combustion releases CO■, fewer trees.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Trees remove CO■ by photosynthesis. Cutting forests reduces uptake. Burning or decay of wood releases stored carbon as CO■, a greenhouse gas that enhances global warming.

MARK BREAKDOWN

Mark 1: Less photosynthesis / CO■ removal.
Mark 2: CO■ released by burning or decay.
Mark 3: Link to greenhouse effect.

COMMON MISTAKES

- Oxygen depletion main cause.
- No link to CO■.

EXAMINER TIP

Deforestation both releases carbon and reduces the sink.

QUESTION 86

Describe how to use a quadrat and random sampling to estimate plant species frequency in a field.

QUESTION TYPE

Topic: Ecology | **Command word:** describe | **Marks:** 2 | **Skill tested:** Sampling

WHAT THE QUESTION IS ASKING

Random coordinates, count hits, repeat.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Divide field into a grid using tape measures. Use random numbers to place quadrats. Record species touching or within each quadrat. Repeat at least ten times and calculate mean frequency or percentage cover.

MARK BREAKDOWN

Mark 1: Random placement of quadrats.

Mark 2: Species recorded with repeats or mean calculated.

COMMON MISTAKES

- Quadrats placed only where plants visible.
- Single sample.

EXAMINER TIP

Random avoids bias towards rich patches.

QUESTION 87

Evaluate the introduction of a non-native predator to control an invasive prey species.

QUESTION TYPE

Topic: Ecology | **Command word:** evaluate | **Marks:** 4 | **Skill tested:** Conservation

WHAT THE QUESTION IS ASKING

May control prey but risk to native species.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The predator may reduce invasive prey protecting native plants. However, the predator might also hunt native species, lack natural enemies and become invasive itself. Monitoring and risk assessment are essential before release.

MARK BREAKDOWN

Mark 1: Possible benefit to ecosystem.

Mark 2: Risk to native species.

Mark 3: Predator may become invasive.

Mark 4: Judgement on caution.

COMMON MISTAKES

- Only benefits listed.
- No evaluation conclusion.

EXAMINER TIP

Biological control can backfire — evaluate both sides.

QUESTION 88

Explain how maintaining biodiversity can help ensure food security.

QUESTION TYPE

Topic: Ecology | **Command word:** explain | **Marks:** 4 | **Skill tested:** Biodiversity

WHAT THE QUESTION IS ASKING

Gene pool for crops, pest control, resilience.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Wild relatives of crops provide alleles for disease resistance or drought tolerance in breeding. Diverse ecosystems support pollinators and natural pest control. Monocultures with low diversity are vulnerable to crop failure.

MARK BREAKDOWN

Mark 1: Wild relatives for breeding.

Mark 2: Pollinators or pest control.

Mark 3: Resilience against disease or famine.

COMMON MISTAKES

- Biodiversity only for tourism.
- No link to food.

EXAMINER TIP

Food security needs genetic resources and ecosystem services.

QUESTION 89

Describe a field investigation to compare biodiversity in two habitats using sweep netting and identification keys. Include how to calculate a simple biodiversity index and improve reliability.

QUESTION TYPE

Topic: Ecology | **Command word:** describe | **Marks:** 6 | **Skill tested:** Fieldwork

WHAT THE QUESTION IS ASKING

Systematic sampling, species count, Simpson or species richness.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

In each habitat, perform ten sweep net samples along transects of equal length at the same time of day. Identify and count species using keys. Record numbers in a table. Calculate species richness or Simpson's index for each habitat. Repeat on another day and compare means. Control weather and sampler skill.

MARK BREAKDOWN

Mark 1: Sweep netting method.

Mark 2: Identification using keys.
Mark 3: Biodiversity index calculated.
Mark 4: Fair comparison between habitats.
Mark 5: Reliability improvement.
Mark 6: Logical sequence.

COMMON MISTAKES

- No identification method.
- Different transect lengths.

EXAMINER TIP

Species richness = number of different species present.

GRADE 7–9 EXTENSION

Higher index generally means higher biodiversity.

QUESTION 90

In a food chain, grass transfers 10 000 kJ to rabbits. Only 10% passes to foxes. Calculate the energy available to foxes.

QUESTION TYPE

Topic: Ecology | **Command word:** calculate | **Marks:** 3 | **Skill tested:** Calculation

WHAT THE QUESTION IS ASKING

Multiply energy by efficiency as decimal.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Energy to foxes = 10 000 × 0.10 = 1000 kJ.

MARK BREAKDOWN

Mark 1: Uses 10% or 0.10.
Mark 2: Correct multiplication set up.
Mark 3: 1000 kJ.

COMMON MISTAKES

- Subtracting 10%.
- Using 90%.

EXAMINER TIP

Energy transfer between trophic levels is roughly 10%.

QUESTION 91

Describe how you would investigate the effect of a pollutant such as acid rain on seed germination using cress seeds on filter paper.

QUESTION TYPE

Topic: Ecology | **Command word:** describe | **Marks:** 4 | **Skill tested:** Required practical — pollution

WHAT THE QUESTION IS ASKING

Different pH solutions, count germination over time.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Place equal numbers of cress seeds on damp filter paper in Petri dishes. Add equal volumes of solutions at different pH values. Keep at same temperature and light. Count germinated seeds daily for seven days. Calculate percentage germination. Repeat with fresh seeds.

MARK BREAKDOWN

Mark 1: pH as independent variable.

Mark 2: Germination counted.

Mark 3: Percentage calculated.

Mark 4: Controls named.

COMMON MISTAKES

- Changing temperature each day.
- No repeat.

EXAMINER TIP

Use sulfuric acid diluted carefully with safety goggles.

Section 2: Required practical skills

QUESTION 92

State what is meant by the independent variable in an investigation.

QUESTION TYPE

Topic: Required practical skills | **Command word:** state | **Marks:** 1 | **Skill tested:** Variables

WHAT THE QUESTION IS ASKING

Define the variable changed deliberately.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The variable that is changed or selected by the investigator.

MARK BREAKDOWN

Mark 1: The variable that is changed or selected by the investigator.

COMMON MISTAKES

- The variable measured.
- A variable kept constant.

EXAMINER TIP

Independent = what you change.

QUESTION 93

Give one hazard symbol meaning in a school laboratory.

QUESTION TYPE

Topic: Required practical skills | **Command word:** give | **Marks:** 1 | **Skill tested:** Safety

WHAT THE QUESTION IS ASKING

Name a hazard and meaning.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Flammable — catches fire easily (or toxic, corrosive, irritant).

MARK BREAKDOWN

Mark 1: Flammable — catches fire easily (or toxic, corrosive, irritant).

COMMON MISTAKES

- Wrong symbol meaning.
- Vague 'danger'.

EXAMINER TIP

Link symbol to specific risk and control.

QUESTION 94

State why repeats are used in experiments.

QUESTION TYPE

Topic: Required practical skills | **Command word:** state | **Marks:** 1 | **Skill tested:** Reliability

WHAT THE QUESTION IS ASKING

Improve reliability and identify anomalies.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

To improve reliability and calculate a mean, and to identify anomalous results.

MARK BREAKDOWN

Mark 1: To improve reliability and calculate a mean, and to identify anomalous results.

COMMON MISTAKES

- To make results different.
- To use more equipment.

EXAMINER TIP

Repeats → mean → more reliable data.

QUESTION 95

Describe how to identify an anomalous result in a data set.

QUESTION TYPE

Topic: Required practical skills | **Command word:** describe | **Marks:** 2 | **Skill tested:** Data handling

WHAT THE QUESTION IS ASKING

Compare to other repeats; exclude from mean.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

An anomalous result is very different from other repeats when conditions were the same. It should be excluded from the mean but recorded, and the experiment repeated if possible.

MARK BREAKDOWN

Mark 1: Very different from other values.

Mark 2: Excluded from mean but recorded.

COMMON MISTAKES

- Deleting without recording.
- Calling any highest value anomalous.

EXAMINER TIP

Do not include anomalies in mean calculations.

QUESTION 96

Explain why a control experiment is used.

QUESTION TYPE

Topic: Required practical skills | **Command word:** explain | **Marks:** 2 | **Skill tested:** Experimental design

WHAT THE QUESTION IS ASKING

Comparison to show effect of independent variable.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

A control provides a baseline where the independent variable is absent or standard, so any change in the dependent variable can be attributed to the factor being tested.

MARK BREAKDOWN

Mark 1: Baseline for comparison.

Mark 2: Shows effect of changed variable.

COMMON MISTAKES

- Control is the variable you change.
- No purpose stated.

EXAMINER TIP

Without a control you cannot prove cause.

QUESTION 97

Compare accuracy and precision in measurements.

QUESTION TYPE

Topic: Required practical skills | **Command word:** compare | **Marks:** 2 | **Skill tested:** Measurement

WHAT THE QUESTION IS ASKING

True value vs repeat closeness.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Accuracy is how close a measurement is to the true value. Precision is how close repeated measurements are to each other, regardless of whether they are correct.

MARK BREAKDOWN

Mark 1: Accuracy: closeness to true value.

Mark 2: Precision: repeat closeness.

COMMON MISTAKES

- Using as synonyms.
- No contrast.

EXAMINER TIP

Precise but inaccurate = consistent wrong answers.

QUESTION 98

Describe how to draw a line graph from experimental data including axes, units and line of best fit.

QUESTION TYPE

Topic: Required practical skills | **Command word:** describe | **Marks:** 3 | **Skill tested:** Graph skills

WHAT THE QUESTION IS ASKING

IV on x, DV on y, scale, points, smooth curve or line.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Plot the independent variable on the x-axis and dependent on the y-axis with labels and units. Use a sensible scale. Plot points as crosses. Draw a smooth curve or straight line of best fit through the trend, not dot-to-dot.

MARK BREAKDOWN

Mark 1: Correct axes and labels.
Mark 2: Units on axes.
Mark 3: Line of best fit not dot-to-dot.

COMMON MISTAKES

- Swapped axes.
- Joining every point regardless of trend.

EXAMINER TIP

Line of best fit shows overall trend.

QUESTION 99

Explain how to convert units from millimetres to micrometres.

QUESTION TYPE

Topic: Required practical skills | **Command word:** explain | **Marks:** 3 | **Skill tested:** Units

WHAT THE QUESTION IS ASKING

Multiply by 1000 for smaller unit.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

There are 1000 µm in 1 mm, so multiply the measurement in mm by 1000 to convert to µm. For example, 0.05 mm = 50 µm.

MARK BREAKDOWN

Mark 1: 1000 µm in 1 mm.

Mark 2: Multiply mm by 1000.

Mark 3: Example conversion.

COMMON MISTAKES

- Divide instead of multiply.
- Using 100 factor.

EXAMINER TIP

Going to smaller units → multiply.

QUESTION 100

A student's graph shows a straight line through the origin with gradient 2. Determine what this tells you about the relationship between the variables.

QUESTION TYPE

Topic: Required practical skills | **Command word:** determine | **Marks:** 4 | **Skill tested:** Graph interpretation

WHAT THE QUESTION IS ASKING

Direct proportionality; DV = 2 × IV.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The dependent variable is directly proportional to the independent variable. When IV doubles, DV doubles. The relationship can be written as DV = 2 × IV.

MARK BREAKDOWN

Mark 1: Direct proportion identified.

Mark 2: Gradient meaning explained.

Mark 3: Example doubling relationship.

Mark 4: Equation or clear statement.

COMMON MISTAKES

- Inverse proportion.
- Gradient ignored.

EXAMINER TIP

Through origin + straight line = direct proportion.

QUESTION 101

Evaluate using mean alone without range or repeat data when reporting an investigation.

QUESTION TYPE

Topic: Required practical skills | **Command word:** evaluate | **Marks:** 4 | **Skill tested:** Data analysis

WHAT THE QUESTION IS ASKING

Mean hides spread and reliability issues.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The mean summarises central tendency but hides how spread out repeats are. Without range or standard deviation, an unreliable mean from widely different repeats may mislead. Reporting range and number of repeats strengthens conclusions.

MARK BREAKDOWN

Mark 1: Mean alone insufficient.

Mark 2: Range or spread matters.

Mark 3: Reliability concern.

Mark 4: Better reporting suggested.

COMMON MISTAKES

- Mean always enough.
- No limitation given.

EXAMINER TIP

Always pair mean with spread when possible.

QUESTION 102

Describe how you would plan, carry out and evaluate an investigation to test whether pH affects amylase activity using starch and iodine. Include risk assessment and improvements.

QUESTION TYPE

Topic: Required practical skills | **Command word:** describe | **Marks:** 6 | **Skill tested:** Enzyme practical planning

WHAT THE QUESTION IS ASKING

Full plan with variables, method, safety and evaluation.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Set up starch solution with amylase at different pH buffers in a water bath at 37 °C. Start timer and sample mixture every 30 s, add to iodine on a spotting tile. Record time until iodine stays orange-brown (starch gone). Control enzyme mass, starch volume and temperature. Risk: allergies or glass — goggles and care with iodine. Improve by triplicate repeats and colorimeter for colour change endpoint.

MARK BREAKDOWN

Mark 1: pH as independent variable.
Mark 2: Iodine test for starch breakdown.
Mark 3: Temperature controlled.
Mark 4: Risk and control.
Mark 5: Evaluation improvement.
Mark 6: Coherent plan.

COMMON MISTAKES

- No endpoint described.
- Changing temperature with pH.

EXAMINER TIP

Orange-brown iodine = no starch = reaction complete.

GRADE 7–9 EXTENSION

Optimum pH for amylase is around neutral pH 7.

QUESTION 103

Five repeat measurements of mass are 2.4, 2.6, 2.5, 2.5 and 2.0 g. Calculate the mean mass excluding the anomalous result.

QUESTION TYPE

Topic: Required practical skills | **Command word:** calculate | **Marks:** 3 | **Skill tested:** Calculation

WHAT THE QUESTION IS ASKING

Exclude 2.0 g, sum others and divide by 4.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Anomaly = 2.0 g. Mean = (2.4 + 2.6 + 2.5 + 2.5) ÷ 4 = 10.0 ÷ 4 = 2.5 g.

MARK BREAKDOWN

Mark 1: Anomalous value identified.
Mark 2: Sum of four values.
Mark 3: Mean = 2.5 g.

COMMON MISTAKES

- Including 2.0 g.
- Dividing by 5 after exclusion.

EXAMINER TIP

State which value you excluded and why.

QUESTION 104

Describe how you would calibrate an eyepiece graticule using a stage micrometer before measuring cell size.

QUESTION TYPE

Topic: Required practical skills | **Command word:** describe | **Marks:** 4 | **Skill tested:** Required practical — microscopy

WHAT THE QUESTION IS ASKING

Align scales, calculate µm per graticule unit.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Place stage micrometer on microscope stage. Count how many eyepiece graticule divisions match a known distance on the micrometer scale (e.g. 0.1 mm). Divide known real distance by graticule divisions to find µm per graticule unit. Use this factor to convert cell graticule measurements to real size.

MARK BREAKDOWN

Mark 1: Both scales viewed at same magnification.

Mark 2: Known real distance used.

Mark 3: Conversion factor calculated.

Mark 4: Apply to cell measurement.

COMMON MISTAKES

- Measuring without calibration.
- Changing magnification after calibration.

EXAMINER TIP

Calibration valid only at the magnification used.

Section 3: Calculation walkthroughs

Calculation 1: Magnification

A chloroplast is 5 mm long on a micrograph. The real length is 2.5 µm. Calculate the magnification.

Formula: Magnification = image size ÷ actual size

Rearrangement: Same formula; convert units first

Substitution: 5 mm = 5000 µm; Magnification = 5000 ÷ 2.5

Working: Magnification = ×2000

Answer: ×2000 (no units)

Magnification has no units. Convert both measurements to the same unit before dividing.

Calculation 2: Percentage change in mass (osmosis)

A potato cylinder had mass 1.20 g before soaking and 1.32 g after. Calculate the percentage change in mass.

Formula: Percentage change = (change ÷ original) × 100

Rearrangement: Change = final – original

Substitution: Change = 1.32 – 1.20 = 0.12 g; (0.12 ÷ 1.20) × 100

Working: Percentage change = 10%

Answer: +10%

Use original mass as denominator. Positive value = mass gain.

Calculation 3: Rate of reaction

In an enzyme experiment, 24 cm^3 of gas was collected in 2 minutes. Calculate the rate in cm^3/min.

Formula: Rate = amount ÷ time

Rearrangement: Rate is subject

Substitution: Rate = 24 ÷ 2

Working: Rate = 12 cm^3/min

Answer: 12 cm^3/min

Rate always includes per unit time.

Calculation 4: Body mass index (BMI)

Calculate the BMI of a person with mass 64 kg and height 1.65 m.

Formula: BMI = mass (kg) ÷ [height (m)]^2

Rearrangement: Square height first

Substitution: BMI = 64 ÷ (1.65^2) = 64 ÷ 2.7225

Working: BMI = 23.5 kg/m^2 (3 s.f.)

Answer: 23.5 kg/m^2

Height must be in metres before squaring.

Calculation 5: Surface area to volume ratio

A cube-shaped organism has side length 2 mm. Calculate its surface area to volume ratio.

Formula: SA = $6s^2$, V = s^3, ratio = SA ÷ V

Rearrangement: Calculate SA and V separately

Substitution: SA = 6 × 2^2 = 24 mm^2; V = 2^3 = 8 mm^3; ratio = 24 ÷ 8

Working: SA:V ratio = 3:1 or 3 mm■^{1}

Answer: 3:1 (or 3 per mm)

Smaller organisms have higher SA:V, aiding exchange.

Calculation 6: Mean and range

Five counts of daisy plants in quadrats are 4, 6, 5, 5 and 10. Calculate the mean and range.

Formula: Mean = sum ÷ n; Range = highest – lowest

Rearrangement: Exclude anomaly if instructed

Substitution: Mean = (4+6+5+5+10) ÷ 5 = 30 ÷ 5; Range = 10 – 4

Working: Mean = 6; Range = 6

Answer: Mean = 6 plants; Range = 6 plants

10 may be anomalous — discuss excluding for refined mean in evaluation.

Calculation 7: Graph interpretation — rate from gradient

A graph of distance moved (cm) against time (min) is a straight line through the origin from 0 to 4 min and 20 cm. Determine the rate of movement.

Formula: Rate = gradient = $\Delta y \div \Delta x$

Rearrangement: Pick two points on the line

Substitution: Rate = 20 cm ÷ 4 min

Working: Rate = 5 cm/min

Answer: 5 cm/min

Straight line through origin indicates constant rate / direct proportion.

Calculation 8: Percentage of population

In a sample of 150 students, 27 are colour-blind recessive homozygotes. Calculate the percentage of the sample with this phenotype.

Formula: Percentage = (part ÷ whole) × 100

Rearrangement: Percentage is subject

Substitution: (27 ÷ 150) × 100

Working: Percentage = 18%

Answer: 18%

Useful in genetic and epidemiological contexts.

Section 4: Required practical walkthroughs

Practical 1: Microscopy: observing and measuring cells

Aim: To prepare a slide of plant epidermis, observe cells under a microscope and estimate their size using a calibrated graticule.

Variables: Independent: specimen (e.g. onion epidermis). Dependent: cell dimensions in µm. Control: same magnification, same staining method.

Method: Peel thin epidermis, mount on slide with iodine stain and coverslip. View under low power then high power. Calibrate eyepiece graticule with stage micrometer at same magnification. Measure width of five cells in graticule units, convert to µm, calculate mean.

Risk assessment: Iodine stains skin — wear goggles and avoid contact. Handle glass slides and coverslips carefully to prevent cuts. Do not use cracked slides.

Results table: Table: cell number, graticule units, real width/µm. Include repeats and mean ± optional range.

Graph guidance: Optional bar chart of mean cell width with error bars from range of repeats.

Conclusion: Onion epidermis cells are rectangular with cell wall, cytoplasm, nucleus and often vacuole visible. Mean width should be consistent at same magnification.

Evaluation: Random error from measuring different cells; systematic error if calibration magnification changes. Improve by measuring more cells and using median if one cell differs greatly.

Common exam questions:

- Why is iodine used?
- Why calibrate the graticule?
- How do you calculate real size from graticule units?

Model answers:

- Iodine stains starch and nuclei, making structures more visible.
- Graticule units are arbitrary; calibration converts them to real µm at that magnification.
- Real size = graticule reading × µm per graticule unit from calibration.

Practical 2: Osmosis in potato cylinders

Aim: To investigate how sugar solution concentration affects water movement into or out of potato tissue.

Variables: Independent: molarity of sugar solution. Dependent: percentage change in mass of potato cylinders. Control: same potato, same dimensions, same time, same temperature.

Method: Cut equal-sized potato cylinders. Blot dry and record initial mass. Place one cylinder in each concentration of sugar solution for 30 minutes. Remove, blot dry, record final mass. Calculate percentage change in mass for each concentration.

Risk assessment: Low risk — wash hands after handling food. Mop spills to prevent slips. Scalpel cuts — cut away from body on tile.

Results table: Table: concentration/mol dm■^{3}, initial mass/g, final mass/g, % change. Expect mass loss in concentrated solutions, gain in pure water.

Graph guidance: Plot percentage change in mass (y) against sugar concentration (x). Draw smooth curve crossing 0% change at approximate cell sap concentration.

Conclusion: Water moves by osmosis from higher water potential to lower. Potato loses mass in hypertonic solutions and gains in hypotonic water.

Evaluation: Surface drying affects mass — blot consistently. Improve by using borer for equal diameter and repeating entire experiment on another day.

Common exam questions:

- Why blot cylinders before weighing?
- What does zero percentage change indicate?
- Which solution is hypertonic to potato cells if mass decreases?

Model answers:

- Removes surface water so measured mass change is due to osmosis in tissue, not surface moisture.
- No net osmosis — solution is isotonic to cell sap at that concentration.
- The more concentrated sugar solution where mass decreased — lower water potential than cell sap.

Practical 3: Investigating the effect of pH on amylase activity

Aim: To find how pH affects the time taken for amylase to break down starch.

Variables: Independent: pH (using buffer solutions). Dependent: time for starch to disappear (iodine test). Control: temperature 37 °C, same enzyme and starch concentrations.

Method: Mix amylase with starch in test tube in water bath at 37 °C with buffer at set pH. Every 30 s remove a drop, mix with iodine on tile. Record time when iodine stays orange-brown. Repeat at different pH values.

Risk assessment: Iodine irritant — goggles and avoid skin contact. Water bath — check electrical safety. Hot water burn risk — use tongs.

Results table: Table: pH, time/s for starch breakdown. Fastest time at optimum pH (~7 for amylase).

Graph guidance: Bar chart or line graph of time (y) against pH (x). Lower time = faster activity.

Conclusion: Enzyme activity depends on pH because extreme pH denatures active site shape. Amylase works fastest near neutral pH.

Evaluation: Colour judgement subjective — use colorimeter or triplicate drops. Improve by fresh enzyme each run and precise buffer pH meter check.

Common exam questions:

- Why use a water bath at 37 °C?
- How do you know starch is gone?
- Why does very low pH slow the reaction?

Model answers:

- Maintains constant optimum temperature for enzyme activity.
- Iodine no longer turns blue-black — starch absent.
- Low pH changes active site shape — enzyme denatured or less effective.

Section 5: Full mini mock paper

Total marks: 50

Question 1 (1 marks)

State the function of the nucleus.

Question 2 (2 marks)

Give one difference between xylem and phloem.

Question 3 (3 marks)

Describe how white blood cells defend against pathogens.

Question 4 (3 marks)

Explain why enzymes are denatured at high temperatures.

Question 5 (3 marks)

An image is 12 mm long and the real object is 6 µm long. Calculate the magnification.

Question 6 (3 marks)

Compare the hormonal and nervous systems.

Question 7 (4 marks)

Describe how natural selection can lead to antibiotic resistance.

Question 8 (3 marks)

Explain how deforestation affects biodiversity and climate.

Question 9 (3 marks)

A person has BMI calculated as 31. Suggest what this indicates and one health implication.

Question 10 (3 marks)

Describe how you would investigate osmosis in potato tissue.

Question 11 (4 marks)

Explain the role of insulin and glucagon in controlling blood glucose.

Question 12 (6 marks)

Evaluate the use of vaccination programmes in controlling disease.

Question 13 (3 marks)

In a species the allele for dark fur (D) is dominant to light fur (d). Cross Dd × dd and explain expected offspring phenotypes.

Question 14 (3 marks)

A quadrat survey gives daisy counts: 3, 4, 4, 5, 14. Calculate mean excluding anomaly and the range of the remaining data.

Question 15 (6 marks)

Describe how you would use a microscope to estimate the size of cheek epithelial cells.

Mini mock — step-by-step walkthrough answers

Answer 1

The nucleus controls cell activities and contains genetic material (DNA).

Mark breakdown:

Mark 1: Controls activities or contains DNA.

Answer 2

Xylem transports water and minerals upward; phloem transports sugars both up and down. Xylem cells are dead with lignin; phloem contains living sieve tubes.

Mark breakdown:

Mark 1: Transport difference.

Mark 2: Structural or directional difference.

Answer 3

Phagocytes engulf pathogens and digest them. Lymphocytes produce antibodies that bind to antigens on pathogens, causing clumping or marking for destruction. Some form memory cells.

Mark breakdown:

Mark 1: Phagocytosis described.

Mark 2: Antibody production.

Mark 3: Memory or antigen binding.

Answer 4

Enzymes have an active site with a specific shape. High temperature breaks bonds holding the tertiary structure, changing active site shape so the substrate no longer fits. The enzyme is denatured and cannot catalyse the reaction.

Mark breakdown:

Mark 1: Active site shape specific.

Mark 2: High temperature changes shape.

Mark 3: Substrate no longer fits / enzyme denatured.

Answer 5

Convert: 12 mm = 12 000 µm. Magnification = 12 000 ÷ 6 = ×2000.

Mark breakdown:

Mark 1: Unit conversion.

Mark 2: Correct formula.

Mark 3: ×2000.

Answer 6

Nervous: electrical impulses, fast, short-lived, via neurones. Hormonal: chemical hormones in blood, slower, longer-lasting, widespread. Nervous good for rapid reflexes; hormones for gradual changes like menstruation.

Mark breakdown:

Mark 1: Nervous features.

Mark 2: Hormonal features.

Mark 3: Clear comparison.

Answer 7

Random mutations produce resistant bacteria. Antibiotics kill susceptible bacteria, leaving resistant ones to reproduce. Over generations allele frequency for resistance increases — evolution.

Mark breakdown:

Mark 1: Variation/mutation.

Mark 2: Selection by antibiotic.

Mark 3: Survivors reproduce.

Mark 4: Population change over time.

Answer 8

Removing forests destroys habitats, reducing species diversity and food webs. Less photosynthesis means less CO■ removed. Burning or decay releases CO■, contributing to greenhouse effect and climate change.

Mark breakdown:

Mark 1: Habitat loss reduces biodiversity.

Mark 2: Less CO■ uptake.

Mark 3: Climate link.

Answer 9

BMI above 30 indicates obesity. Health implications include increased risk of type 2 diabetes, cardiovascular disease or joint problems.

Mark breakdown:

Mark 1: Obesity identified.

Mark 2: Valid health implication.

Mark 3: Linked explanation.

Answer 10

Cut equal potato cylinders, record mass, place in different sugar concentrations for fixed time at same temperature. Blot and reweigh. Calculate percentage change. Plot mass change against concentration.

Mark breakdown:

Mark 1: Method with cylinders and solutions.

Mark 2: Mass change measured.

Mark 3: Controls named.

Answer 11

After a meal glucose rises; pancreas releases insulin causing cells to take up glucose and store as glycogen, lowering blood glucose. If glucose falls too low, glucagon is released, converting glycogen to glucose in the liver.

Mark breakdown:

Mark 1: Insulin lowers glucose.

Mark 2: Glucagon raises glucose.

Mark 3: Glycogen storage/release.

Mark 4: Pancreas source.

Answer 12

Vaccination stimulates memory cells without severe disease, protecting individuals. Herd immunity reduces spread, protecting vulnerable people. Limitations: not 100% effective, some cannot vaccinate, booster needed, rare side effects. Overall public health benefit outweighs risks for most programmes.

Mark breakdown:

Mark 1: Individual protection.

Mark 2: Herd immunity.

Mark 3: Limitation stated.

Mark 4: Side effects or access issues.

Mark 5: Balanced judgement.

Mark 6: Coherent evaluation.

Answer 13

Gametes: D or d from Dd parent; d from dd parent. Offspring: Dd or dd in 1:1 ratio — half dark fur (Dd), half light fur (dd).

Mark breakdown:

Mark 1: Gametes identified.

Mark 2: 1:1 phenotype ratio.

Mark 3: Link to dominant/recessive.

Answer 14

14 is anomalous. Mean of 3,4,4,5 = 16÷4 = 4. Range = 5 – 3 = 2.

Mark breakdown:

Mark 1: Anomaly excluded.

Mark 2: Mean = 4.

Mark 3: Range = 2.

Answer 15

Scrape cheek lining gently with cotton bud, smear on slide, add methylene blue stain and coverslip. Calibrate graticule with micrometer at high power. Measure cell length in graticule units for ten cells, convert to µm, calculate mean. Repeat for reliability.

Mark breakdown:

Mark 1: Safe sampling and staining.

Mark 2: Calibration at set magnification.

Mark 3: Multiple measurements and mean.

Mark 4: Reliability or safety.

Mark 5: Units in µm.

Mark 6: Logical sequence.